Aim

To design and simulate a digital stopwatch system using the 8051 microcontroller with start, stop, reset, and lap features, using Timer0 interrupt in Keil µVision.

Objectives

- To understand the working of **8051 timers**
- To implement **interrupt-based time counting**
- To simulate real-world stopwatch functionality
- To store and display **lap time values**

Introduction

A stopwatch is a time-measuring device used in sports, experiments, and embedded systems. In microcontroller-based systems, timers are used to generate precise delays and track elapsed time.

In this case study, the **8051 microcontroller** is used along with **Timer0 in Mode 1** to generate a **1 millisecond interrupt**, which is used to calculate seconds. The system is simulated in **Keil µVision** without any external hardware.

System Overview

The system consists of:

- Timer0 configured in **16-bit mode**
- Interrupt service routine (ISR) for time counting
- Input controls (Start, Stop, Reset, Lap) using port pins
- Variables to store current time and lap time

Components Used

Component	Description
8051 Microcontroller	AT89C51
Timer0	Used for time generation
Keil µVision	Simulation software
Port Pins (P1.0–P1.2)	Input controls

Working Principle

- Timer0 generates an interrupt every **1 millisecond**
- A counter variable increments for each interrupt
- After 1000 counts → 1 second is recorded
- The stopwatch operates based on control inputs:

- Start/Stop toggles counting
- Reset clears all values
- Lap stores the current time

Algorithm

1. Initialize Timer0 in Mode 1
2. Enable interrupts (EA, ET0)
3. Start Timer (TR0 = 1)
4. On each interrupt:
    - Increment millisecond counter
    - If 1000 ms → increment seconds
5. Check inputs:
    - Start → toggle running state
    - Reset → clear counters
    - Lap → store current time
6. Repeat continuously

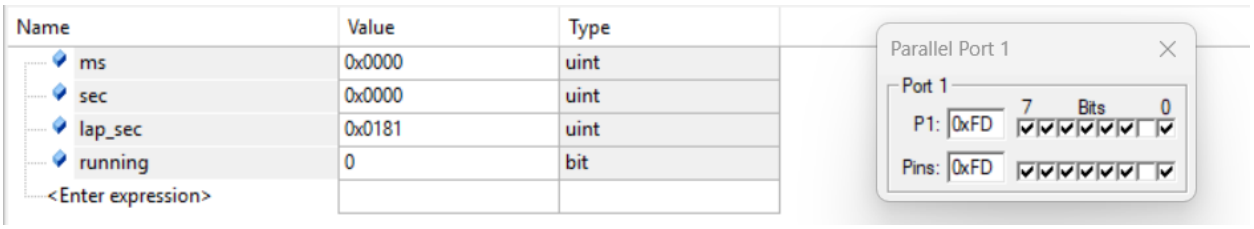
Program Code

```
#include <reg51.h>
sbit START = P1^0;
sbit RESET = P1^1;
sbit LAP   = P1^2;
unsigned int ms = 0;
unsigned int sec = 0;
unsigned int lap_sec = 0;
bit running = 0;
vod timer0_ISR(void) interrupt 1
{
   TH0 = 0xFC;
   TL0 = 0x66;
   if(running)
   {
      ms++;
      if(ms >= 1000)
      {
         ms = 0;
         sec++;
      }
   }
}
void Timer0_Init()
{
   TMOD = 0x01;
   TH0 = 0xFC;
```

```
    TL0 = 0x66;
    ET0 = 1;
    EA  = 1;
    TR0 = 1;
}
void main()
{
    Timer0_Init();
    while(1)
    {
        if(START == 0)
        {
            running = !running;
            while(START == 0);
        }
        if(RESET == 0)
        {
            ms = 0;
            sec = 0;
            while(RESET == 0);
        }

        if(LAP == 0)
        {
            lap_sec = sec;
            while(LAP == 0);
        }
    }
}
```

P1.0 starts



P1.1 resets

P1.0 reselect the same port restart

Simulation Procedure

1. Create a new project in Keil µVision
2. Select **AT89C51 microcontroller**
3. Add the C program file
4. Build the project
5. Start Debug Session
6. Open:
    - I/O Port 1
    - Watch Window
7. Run the program
8. Simulate inputs by toggling port pins

Results

- Stopwatch counts time accurately in seconds
- Start/Stop functionality works correctly
- Reset clears all values
- Lap feature successfully stores current time

- All operations verified in simulation

Advantages

- High timing accuracy using interrupts
- No need for external hardware
- Efficient CPU utilization
- Easy to extend (LCD, UART, etc.)

Limitations

- No physical display (simulation only)
- Limited to seconds (no minutes/hours)
- Button debounce handled minimally

Applications

- Sports timing systems
- Industrial timers
- Embedded system experiments
- Digital clocks

Conclusion

The digital stopwatch with lap feature was successfully designed and simulated using the 8051 microcontroller. The use of Timer0 interrupt ensures accurate time tracking, while control inputs enable real-time interaction. This case study demonstrates practical implementation of timers, interrupts, and embedded system design concepts.